

VACF-to-MSD Reconstruction Specification

GK4 after H0: Projected Correlation-to-Displacement Diagnostic

mdstats

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Purpose and status

This document specifies the H0-hardened

`reconstruct_msd_from_vacf(...)`

implementation in `mdstats/analysis/vacf_transport.py`. It reconstructs the mean-square displacement implied by a stored physical self VACF. It is a consistency diagnostic; direct position-based `compute_msd()` remains primary.

Borrowed theory

For a stationary trajectory projected onto an orthonormal subspace B ,

$$B[\mathbf{r}(t) - \mathbf{r}(0)] = \int_0^t B \mathbf{v}(s) ds.$$

The correlation-to-displacement identity is

$$M_B(t) = 2 \int_0^t (t - \tau) C_B(\tau) d\tau,$$

with

$$C_B(\tau) = \text{tr} [BC(\tau)B^T].$$

The equivalent two-moment implementation is

$$I_0(t) = \int_0^t C_B(\tau) d\tau, \quad I_1(t) = \int_0^t \tau C_B(\tau) d\tau,$$

$$M_B(t) = 2[tI_0(t) - I_1(t)].$$

This identity belongs to the Einstein/Green-Kubo/Helfand transport framework [1-4]. The two-cumulative-integral sampled implementation, explicit subspace API, and provenance schema are `mdstats` design.

Public function

```
def reconstruct_msd_from_vacf(  
    vacf: VACFResult,  
    *,
```

```

axes: Sequence[Literal["x", "y", "z"]] | None = None,
projection_basis: ArrayLike | None = None,
component: Literal["scalar", "x", "y", "z"] = "scalar",
maximum_time_ps: float | None = None,
integration: Literal["trapezoid"] = "trapezoid",
) -> VACFMSDResult:
    ...

```

axes or projection_basis defines the physical subspace. The legacy component adapter is accepted only when no explicit subspace is supplied. Default scalar means full 3D; x/y/z means that one axis.

The equal-positive-weight, lag validation, tensor requirement, and truncation rules are identical to integrate_vacf_to_diffusion().

No dimensional divisor is applied: the result is the total projected MSD $M_B(t)$, not D_B .

Result contract

```

@dataclass(frozen=True, slots=True)
class VACFMSDResult:
    lag_times: NDArray[np.float64]
    reconstructed_msd_a2: NDArray[np.float64]
    physical_vacf_a2_per_ps2: NDArray[np.float64]
    cumulative_vacf_a2_per_ps: NDArray[np.float64]
    cumulative_time_weighted_vacf_a2: NDArray[np.float64]
    component: str
    weighting: str
    integration: str
    metadata: Mapping[str, Any]
    dimensions: int
    projection_basis: NDArray[np.float64]
    projection_labels: tuple[str, ...] | None
    signature: DynamicsInputSignature | None

```

The constructor requires:

- every numerical array has shape (L,) and finite entries;
- lag time begins at zero and is strictly increasing;
- reconstructed MSD and both cumulative moments begin at zero;
- the basis rank equals dimensions;
- the component label agrees with a single Cartesian basis when applicable;
- the stored signature subspace agrees with the result; and
- both cumulative integrals and the reconstructed MSD reproduce the defining trapezoidal identities within strict tolerance.

All arrays and nested metadata are deeply immutable.

Algorithm

1. Resolve and validate the physical subspace.
2. Validate physical equal-positive self weighting.
3. Retain the accepted lag prefix.
4. Project the source VACF onto the subspace.
5. Compute I_0 and I_1 by cumulative trapezoidal integration.
6. Compute $2(tI_0 - I_1)$ and force the exact zero-lag value to zero.
7. Propagate the source signature with the selected subspace.
8. Validate and freeze the result.

The implementation is $O(L)$ after the projected VACF has been assembled.

Interpretation limits

Finite-record reconstructed and direct MSD curves need not match exactly because:

- the VACF assumes stationary time-origin averaging;
- direct fixed-origin MSD is a different estimator;
- noisy long-lag correlations accumulate under integration;
- finite-difference velocities may attenuate high frequencies; and
- source and direct estimators can use different finite-origin weighting.

These limitations remain visible in source metadata. Reconstruction does not replace direct MSD, choose a diffusion plateau, or classify a dynamical regime.

Required tests

- zero and constant projected VACFs reproduce analytic sampled identities;
- full-3D and Cartesian legacy calls preserve pre-H0 values;
- axis-subset additivity holds;
- rotated projections include tensor cross terms;
- missing tensor data for a rotated basis raises;
- uniform weighting and explicit-equal weighting agree;
- mass and nonuniform weights raise;
- truncation uses existing lag samples only;
- all stored identities are constructor-validated;
- signatures propagate with the selected subspace; and
- every array and nested metadata object is immutable.

References

- [1] A. Einstein, *Ann. Phys.* **322**, 549-560 (1905). DOI: 10.1002/andp.19053220806.
- [2] M. S. Green, *J. Chem. Phys.* **22**, 398-413 (1954). DOI: 10.1063/1.1740082.
- [3] R. Kubo, *J. Phys. Soc. Jpn.* **12**, 570-586 (1957). DOI: 10.1143/JPSJ.12.570.
- [4] E. Helfand, *Phys. Rev.* **119**, 1-9 (1960). DOI: 10.1103/PhysRev.119.1.